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Author Name(s)

Affiliation(s)

Correspondence: corresponding.author@institution.edu

Abstract

Cellular neighborhoods and microenvironments (MEs) summarize the local tissue contexts in which individual cells reside. In spatial proteomics, these labels are commonly constructed by a data-driven pipeline that nevertheless requires investigator decisions at both ends: cell phenotypes are manually consolidated or gated, a neighborhood scale and number of clusters are selected, and the resulting clusters are interpreted and named. Consequently, a carefully curated ME ontology is often used only within the cohort in which it was created. We developed MultiScale CenterQuery-CELLME, a supervised, patient-inductive model that converts an existing expert-defined ME ontology into a reusable cell-level predictor. For each center cell, CELLME uses only semantic cell types and spatial relationships derived from coordinates. A local branch applies two graph isomorphism network layers and center- and hop-conditioned attention to a two-hop subgraph of a within-section 6-nearest-neighbor graph. A second branch attends over five nested neighborhoods of 4, 7, 11, 20, and 33 cells, and updates the local representation through a zero-initialized scalar-gated residual.

Across colorectal cancer (CRC), diffuse large B-cell lymphoma (DLBCL), and non-small cell lung cancer (NSCLC), comprising 2,883,030 cells from 253 patients, CELLME accurately transferred native ME labels to completely held-out patients. It ranked first at every training proportion in CRC and NSCLC, with mean macro-F1 values of 0.9919 and 0.9149 across 15 split-proportion combinations, and was comparable to the strongest local-composition baseline in DLBCL. Frozen models trained on only 20% of either the CRC or NSCLC source cohort produced non-random zero-shot partitions in an independent CRC CODEX cohort. With only seven target training patients and a linear target head, held-out macro-F1 reached 0.6147 and 0.6989, respectively. In held-out CRC patients, predicted MEs reproduced local ecological prototypes, tumor-infiltrating lymphocyte analyses, and exploratory clinical stratification; in an individual tissue region, assignments formed coherent spatial domains and agreed with protein programs withheld from the model. Thus, CELLME turns cohort-specific expert annotation into a transferable spatial representation while preserving explicit uncertainty and dataset-native biological meaning.

Keywords: spatial proteomics; cellular neighborhood; tumor microenvironment; graph neural network; transfer learning; imaging mass cytometry

1 Introduction

Highly multiplexed tissue imaging resolves the identities and positions of thousands to millions of cells while preserving tissue architecture. This has motivated the use of cellular neighborhoods (CNs), recurrent cellular neighborhoods (RCNs), tissue cellular neighborhoods, and microenvironments (MEs) as local ecological units that complement cell identity: a CD8 T cell, for example, may occupy a tumor core, stromal band, immune aggregate, or invasive boundary, with different biological implications in each context [1, 2, 3]. These spatial units have supported mechanistic and clinical analyses in colorectal cancer (CRC), diffuse large B-cell lymphoma (DLBCL), and non-small cell lung cancer (NSCLC) [4, 5, 6].

Table 1: Data-driven and investigator-defined components of the three native ME ontologies. The labels used for supervision were released by the original studies and are treated as reference ontologies rather than absolute biological ground truth.

Dataset	Data-driven component	Investigator-defined component
CRC	Local composition over ten cell types followed by K -means	Ten-neighbor window, 40- μ m truncation, $K = 8$, and post hoc interpretation and naming
DLBCL	Phenograph clusters [9]; 20-cell composition transformed by spatial LDA and clustered by K -means	Manual merging/splitting and thresholds for CD4, CD8, CD20, CD163, PD-1, PD-L1, and CD96; 19 neighbors, ten motifs, and $K = 10$
NSCLC	Window-frequency vectors followed by MiniBatch K -means	Pooled versus histology-specific analysis, 10NN versus 3NN windows, and 10 versus 30 CNs were varied within the study

The most common CN/RCN discovery workflow is data driven but not decision free. Investigators first define a semantic cell-type vocabulary, often by merging or splitting unsupervised clusters, applying marker thresholds, or consulting pathology. They then choose a neighborhood scale, compress each neighborhood to a cell-type count or frequency vector, select an unsupervised model and cluster number, and interpret the resulting mean compositions to assign biological names. The three source studies used here illustrate this dependence (Table 1). CRC used up to ten nearest neighbors with a 40- μ m cutoff and $K = 8$; DLBCL combined Phenograph, manual phenotype refinement, 20-cell compositions, latent Dirichlet allocation, and $K = 10$; and NSCLC explored different pooled or separate analyses, neighborhood sizes, and cluster numbers [4, 5, 6]. The resulting labels are valuable expert-defined ontologies, but they are usually recreated and reinterpreted for each new dataset.

Graph and deep-learning methods can represent tissue architecture beyond fixed composition vectors. SPACE-GM models local cellular graphs to learn disease-associated tissue features, whereas CytoCommunity uses graph pooling for unsupervised or sample-label-supervised discovery of tissue neighborhoods [7, 8]. These approaches address important discovery and outcome-prediction problems. A distinct practical problem remains: once a study has already invested in a cell-level ME ontology, how can that expert knowledge be transferred reproducibly to unseen patients and reused when a new cohort arrives?

We formulate this problem as center-cell classification. Given semantic cell types and coordinate-defined spatial relationships, the objective is to predict the native ME assigned to each center cell. We introduce MultiScale CenterQuery-CELLME, hereafter CELLME, which combines a center-aware local graph representation with a label-free, multiscale cell-type context. The model never reads protein intensities, raw coordinate values, patient identifiers, clinical outcomes, or neighboring ME labels. It therefore learns how cell types are spatially organized under an existing ontology rather than memorizing molecular intensities or cohort identifiers.

We evaluated CELLME in four stages. First, we performed 585 matched patient-held-out benchmark runs across CRC, DLBCL, and NSCLC. Second, we froze models trained on small source cohorts and transferred them to an independent CRC CODEX dataset with different cell-type and ME ontologies, testing both zero-shot partition alignment and seven-patient low-shot adaptation. Third, we replaced native CRC neighborhoods with predicted MEs in downstream analyses modeled on the source publication. Fourth, we examined a held-out tissue region for spatial continuity, local cell composition, protein-program coherence, and boundary-localized uncertainty. Together, these experiments test whether a cohort-specific

annotation scheme can be converted into a reusable spatial model without overstating the native labels as universal ground truth.

2 Model

CELLME predicts one ME distribution for each center cell (Fig. 1). The local branch begins with a Euclidean 6-nearest-neighbor (6NN) graph constructed independently within each slice or core. The graph is symmetrized by bidirectional union: if either cell selects the other, both directed edges are retained. For a center cell i , the model extracts the induced subgraph containing all nodes within two graph hops. Each node v is represented by its one-hot semantic cell type and log-transformed degree in that center-specific subgraph,

$$\mathbf{x}_v = \left[\text{onehot}(t_v) \parallel \log(d_v^{\text{induced}} + 2) \right]. \quad (1)$$

Two 128-dimensional graph isomorphism network (GIN) layers propagate local information [10]. At each layer, a separate attention readout scores every node using its state, the current center state, their elementwise interaction, and a learned 0/1/2-hop embedding. The two 128-dimensional center-conditioned readouts are concatenated to form $\mathbf{z}_{\text{local}} \in \mathbb{R}^{256}$.

The context branch captures spatial scale without constructing a second message-passing graph. For the same center, it forms a root-first ordered sequence containing the center and its 32 nearest cells within the slice. Five nested, self-inclusive prefixes $K \in \{4, 7, 11, 20, 33\}$ are pooled using shared attention conditioned on the center cell type and neighbor rank. These windows are cell counts, not fixed radii or disjoint rings. A second attention module uses $\mathbf{z}_{\text{local}}$ as a query to combine the five 64-dimensional scale-specific pools into a context vector $\mathbf{c}_i \in \mathbb{R}^{64}$.

Finally, the model concatenates local and context evidence and learns both a 256-dimensional correction Δ_i and a scalar gate g_i :

$$\mathbf{f}_i = [\mathbf{z}_{\text{local},i} \parallel \mathbf{c}_i], \quad (2)$$

$$g_i = \sigma(\text{MLP}_g(\mathbf{f}_i)), \quad (3)$$

$$\mathbf{z}_{\text{final},i} = \mathbf{z}_{\text{local},i} + g_i \Delta_i. \quad (4)$$

The last layer of the correction network is initialized to zero and the gate bias to -3 , so training starts exactly at $\mathbf{z}_{\text{final}} = \mathbf{z}_{\text{local}}$. All branches are then optimized jointly. A $256 \rightarrow 64 \rightarrow 64 \rightarrow C$ classifier returns a probability distribution over the dataset-specific native ME classes. The CRC, DLBCL, and NSCLC variants contain 265,285, 266,759, and 266,502 trainable parameters, respectively.

Training and prediction. The complete network is trained end-to-end from scratch using unweighted multiclass cross-entropy. Optimization uses Adam [21] with learning rate 10^{-3} , weight decay 10^{-4} , gradient clipping at 5.0, and an effective batch size of 2,048. Training runs for at most 100 epochs with early-stopping patience of 15; the checkpoint is selected by validation macro-F1, with validation cross-entropy and then earlier epoch used as tie breakers. Train, validation, and test patients are mutually exclusive. The model contains no class weighting, auxiliary objective, contrastive loss, neighboring-ME context, clinical prediction head, or pretrained frozen backbone.

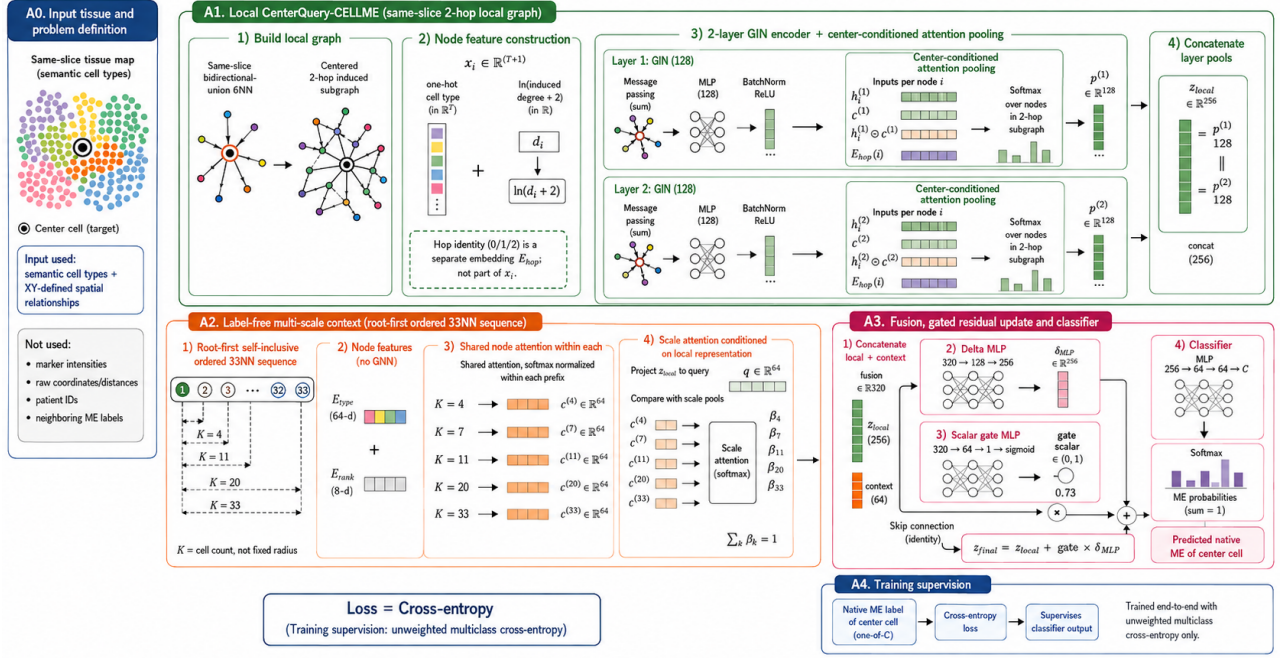


Figure 1: MultiScale CenterQuery-CELLME architecture. For each center cell, the local branch encodes a centered two-hop induced subgraph of a within-slice bidirectional-union 6NN graph using two 128-dimensional GIN layers and layer-wise attention conditioned on the center state and 0/1/2-hop identity. The resulting 256-dimensional local representation conditions attention over five nested, self-inclusive windows of 4, 7, 11, 20, and 33 cells. A zero-initialized scalar-gated residual updates the local representation before classification into the dataset-native ME ontology. Protein intensities, raw coordinate or distance values, clinical covariates, and neighboring ME labels are not model inputs.

3 Results

3.1 Patient-held-out ME prediction across three datasets

Experimental setting. We first tested whether a model trained on a subset of patients could label every cell in completely unseen patients from the same cohort. The benchmark included 2,883,030 cells across 253 patients and 571 slices or cores (Table 2). The three native ontologies differed in technology, disease, phenotype vocabulary, and number of output classes. The NSCLC analysis excluded the source study’s undefined CN3 class, leaving nine semantic MEs.

Table 2: Spatial proteomics cohorts used for in-domain benchmarking.

Cohort	Platform	Patients	Slices/cores	Cells	Cell types	MEs
Stage III CRC	IMC	52	170	903,125	10	8
Helsinki DLBCL	12-plex mIF	99	197	1,218,756	17	10
NSCLC LUAD/LUSC	IMC	102	204	761,149	16	9

For each dataset, we created three fixed patient-level splits. Validation and test patients were held constant within a split, while the training set increased in nested proportions from 20% to 40% of the cohort. All methods used the same center cells, graph construction, cell-type inputs, and test sets.

The comparison included logistic regression, random forest [15], and XGBoost [16] on local-composition vectors; a MiniBatch- K -means phenotype-to-CN approximation [17]; DeepSets [14]; GCN, GraphSAGE, GATv2, and GIN [11, 12, 13, 10]; three CELLME architectural variants; and the final multiscale model. A SPACE-GM-ME adapter [7] was additionally evaluated in CRC only. The primary metric was cell-level macro-F1, which weights each native ME equally.

Table 3: Patient counts in the nested in-domain benchmark splits.

Dataset	20% train	25%	30%	35%	40%	Validation	Test
CRC	10	13	16	18	21	10	10
DLBCL	20	25	30	35	40	20	20
NSCLC	20	26	31	36	41	20	20

Three split seeds (1401–1403) and paired model seeds (2401–2403) were used. For each seed, 50,000 deterministic patient permutations were screened; candidates lacking an ME class in train, validation, or test were rejected, and the valid split with the smallest summed Jensen–Shannon divergence [18] from the full-cohort ME distribution was retained. The 13-method benchmark comprised $13 \times 3 \times 5 \times 3 = 585$ matched runs. Each point in Fig. 2 is the mean test macro-F1 across the three patient splits, and its error bar is the sample standard deviation.

Results and interpretation. In CRC, CELLME achieved macro-F1 values of 0.9921 ± 0.0017 , 0.9921 ± 0.0010 , 0.9926 ± 0.0021 , 0.9929 ± 0.0018 , and 0.9898 ± 0.0083 at the five training proportions (Fig. 2). Across all 15 split–proportion combinations, its mean was 0.9919 ± 0.0036 . The local CenterQuery variant reached 0.8660 ± 0.0035 , indicating that the multiscale residual contributed substantially beyond center-conditioned graph pooling. CELLME also outperformed the CRC-only SPACE-GM-ME adapter (0.8493 ± 0.0137) in all 15 paired settings (two-sided paired Wilcoxon $p = 6.10 \times 10^{-5}$). Notably, ten of 52 training patients were sufficient for a mean macro-F1 of 0.9921.

DLBCL was more variable across patient splits. CELLME achieved 0.8450 ± 0.0429 , 0.8482 ± 0.0832 , 0.8571 ± 0.0460 , 0.8623 ± 0.0210 , and 0.8626 ± 0.0958 as the training proportion increased. Its overall mean across 15 settings was 0.8550 ± 0.0546 , compared with 0.8586 ± 0.0055 for XGBoost-Local, and CELLME won 8 of 15 paired tasks. It ranked first at the 35% and 40% training proportions. These results support parity with the strongest local-composition baseline rather than uniform dominance. They also suggest that the DLBCL RCN labels are strongly explained by local phenotype composition and that neural performance is more sensitive to which patients are observed.

In NSCLC, CELLME ranked first at all five training proportions, attaining 0.9132 ± 0.0061 , 0.9114 ± 0.0013 , 0.9154 ± 0.0042 , 0.9184 ± 0.0047 , and 0.9160 ± 0.0053 . The overall mean was 0.9149 ± 0.0046 , compared with 0.8428 ± 0.0028 for CenterQuery-CELLME; the multiscale model won all 15 paired settings. Performance was already 0.9132 with 20 of 102 training patients and remained on a stable plateau. Thus, across distinct native ontologies, CELLME transferred the source annotation rules to unseen patients with high data efficiency, while the more heterogeneous DLBCL cohort exposed a meaningful limit.

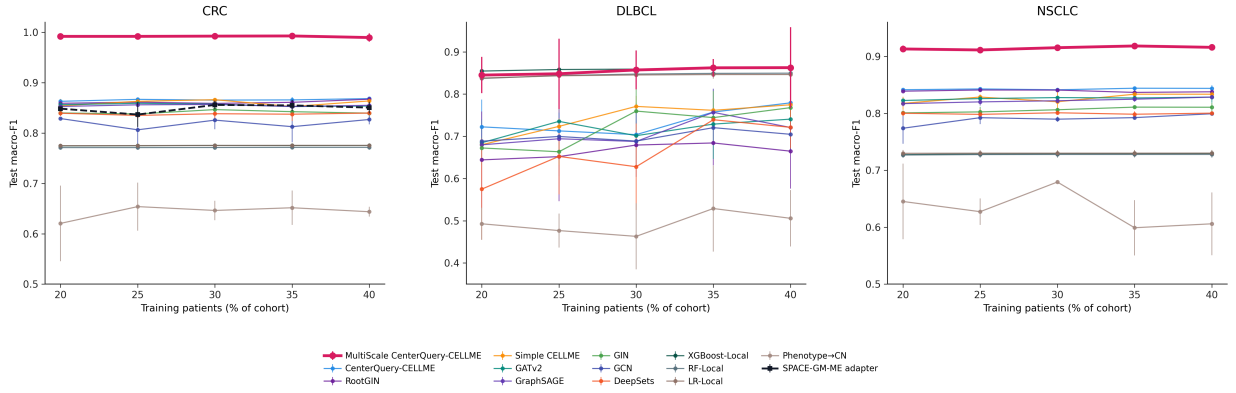


Figure 2: Patient-held-out ME prediction across three spatial proteomics cohorts. Benchmark performance for CRC, DLBCL, and NSCLC across nested patient-level training proportions. Each point represents the mean test macro-F1 across the three fixed patient splits, and error bars denote the sample standard deviation.

3.2 Cross-domain transfer to an independent CRC CODEX cohort

Experimental setting. In-domain prediction evaluates transfer among patients who share a platform and ontology. We next asked whether source-trained models remained useful in an independent advanced-CRC CODEX cohort from Schürch et al. [1]. After excluding cells labeled as dirt or undefined, the target dataset contained 243,188 cells from 35 patients and 140 regions of interest (ROIs), with 27 target cell types and nine target-native MEs. This target differed from both sources in patients, platform, cell-type vocabulary, and ME ontology.

We evaluated two transfer paths: stage III CRC IMC to CRC CODEX, representing same-disease, cross-cohort, cross-platform transfer; and NSCLC IMC to CRC CODEX, representing cross-disease and cross-platform transfer. Each source used the three CELLME models from the 20% patient-training setting (10 of 52 source patients for CRC and 20 of 102 for NSCLC). Target cell types were mapped to the corresponding source vocabulary through a frozen semantic adapter defined without target ME labels, outcomes, or transfer metrics. All source weights were then frozen. Zero-shot predictions retained the source ontology and were evaluated by label-permutation-invariant ARI and NMI. For low-shot transfer, a new nine-class linear head was trained on frozen 256-dimensional representations using seven target patients, selected on seven validation patients, and tested on seven patients in each of three repeated holdouts; the remaining 14 patients were unused.

Results and interpretation. In the zero-shot setting, each model retained its source output ontology, so source class names were not forced into a one-to-one correspondence with the nine target classes. We therefore evaluated label-permutation-invariant adjusted Rand index (ARI) and normalized mutual information (NMI) [19, 20], not target-class F1. Across all target cells, CRC-source predictions achieved $\text{ARI} = 0.2597$ and $\text{NMI} = 0.4300$, while NSCLC-source predictions achieved $\text{ARI} = 0.2797$ and $\text{NMI} = 0.5014$ (Fig. 3a). Across the 140 ROIs, mean NMI was 0.3642 ± 0.1393 for the CRC source and 0.4088 ± 0.1459 for the NSCLC source (Fig. 3b). Transfer was heterogeneous but was not driven by only a few ROIs. In the jointly selected high-alignment ROI reg030_B, the two source ontologies recovered major target spatial domains despite differing class numbers and names (Fig. 3c).

We then tested whether a small amount of target annotation could align the frozen representation to the target-native ontology. For each of three repeated patient holdouts, only seven target patients trained a new nine-class linear head on the frozen 256-dimensional CELLME representation; seven patients were

used for validation and seven for testing, while the remaining 14 were unused. No CELLME weight was fine-tuned. Held-out macro-F1 was 0.6147 ± 0.0143 for CRC-source features and 0.6989 ± 0.0324 for NSCLC-source features (Fig. 3d). Both sources recovered Bulk tumor, Follicle, T-cell-enriched, and Tumor-boundary MEs well; the NSCLC source was particularly stronger for Granulocyte-enriched and Macrophage-enriched MEs, whereas Immune-infiltrated stroma and Vascularized smooth muscle remained difficult (Fig. 3e).

For each held-out target patient, we also compared the nine-component native abundance vector with the mean predicted probability vector. The median within-patient Pearson correlation, averaged across the three splits, was 0.8439 for CRC-source features and 0.9587 for NSCLC-source features (Fig. 3f). The stronger NSCLC transfer in this one target cohort should not be interpreted as a general rule that cross-disease transfer exceeds same-disease transfer. Rather, it demonstrates that the learned spatial representation contains reusable organization principles that can cross both cancer and platform boundaries.

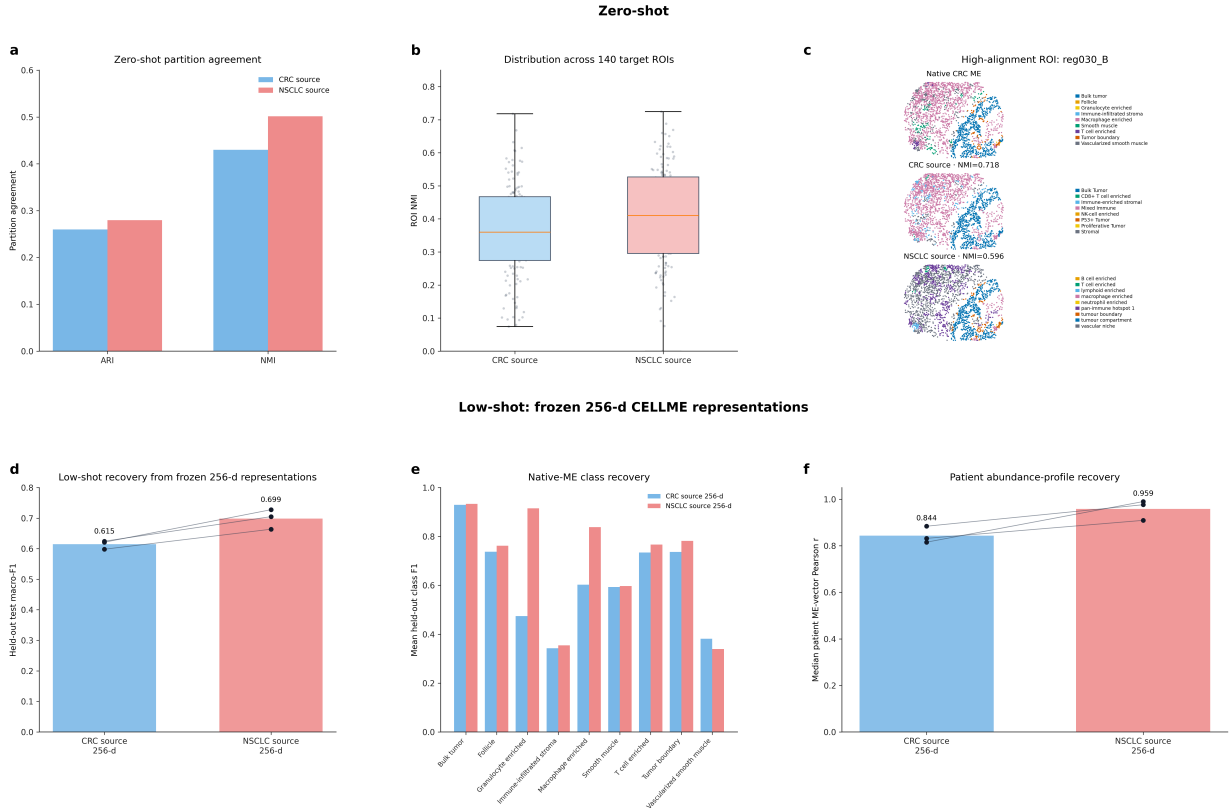


Figure 3: Cross-domain transfer to an independent CRC CODEX cohort. Frozen CELLME models trained with 20% of the patients from either stage III CRC IMC or NSCLC IMC were transferred to 243,188 retained cells across 140 ROIs from 35 independent CRC patients. Target cell types were mapped to each source vocabulary using a frozen semantic adapter that did not use target ME labels. **a**, Zero-shot agreement between the ensemble source-ontology partition and the nine target-native MEs, quantified by ARI and NMI. **b**, ROI-level NMI distributions; points denote ROIs. **c**, Target-native and source-predicted partitions in the jointly selected high-alignment ROI reg030_B (2,089 cells). Colors are ontology specific and do not imply name correspondence. **d**, Target-native macro-F1 after fitting only a nine-class linear head on frozen 256-dimensional representations using seven target training patients. Bars are means across three paired patient holdouts; points and lines show matched splits. **e**, Mean held-out per-class F1. **f**, Recovery of patient-level ME abundance profiles, measured by the median within-patient Pearson correlation across the nine components.

3.3 Reproduction of downstream CRC analyses with predicted MEs

Experimental setting. High label agreement is useful only if predicted compartments support the analyses for which the native ontology was created. We therefore reconstructed seven analyses from the CRC source study [4], replacing native CNs with CELLME predictions in strictly held-out patients (Fig. 4). A model trained on 18 CRC patients (35%, split 00), selected on ten validation patients, and frozen at epoch 23 was applied once to ten disjoint test patients. Full-test performance across 178,986 cells was 0.99296 accuracy, 0.99136 macro-F1, and 0.99118 patient-balanced macro-F1. Downstream analyses excluded three control ROIs and retained 168,295 cells from 35 cancer ROIs, comprising four MSI-H and six microsatellite-stable (MSS) patients.

Local composition used the CRC source definition of up to ten non-self neighbors within 40 μm . Lymphocytes were CD4 T, CD8 T, or B cells; predicted tumor MEs were Bulk Tumor, P53+ Tumor, and Proliferative Tumor; predicted stromal MEs were Stromal and Immune-enriched stromal. Thus, iTIL and sTIL were the lymphocyte fractions inside predicted tumor and stromal compartments, respectively. MSI and recurrence comparisons used exact patient-label permutations rather than treating ROIs as independent samples. The six-patient MSS survival analysis used an outcome-blind Otsu threshold [22] on predicted sTIL followed by Kaplan–Meier and log-rank analysis [23, 24].

Results and interpretation. Using the source study’s analysis window—up to ten non-self neighbors within 40 μm —predicted MEs recovered distinct local ecological prototypes (Fig. 4a). Stromal, Bulk Tumor, Proliferative Tumor, and P53+ Tumor neighborhoods were enriched for their corresponding stromal or tumor phenotypes. Immune-enriched stromal combined stromal and immune populations, CD8+ T-cell-enriched neighborhoods were enriched for CD8 T cells, Mixed Immune neighborhoods contained several immune lineages, and NK-cell-enriched neighborhoods preferentially contained NK cells. Because the heat map was column-wise z -scored across MEs, it reports relative enrichment rather than raw proportions.

Patient-level predicted-ME composition was heterogeneous across the ten held-out patients (Fig. 4b,c). We next used predicted MEs to define intratumoral and stromal tumor-infiltrating lymphocyte fractions. Lymphocytes were CD4 T, CD8 T, or B cells; predicted tumor MEs were Bulk Tumor, P53+ Tumor, and Proliferative Tumor; predicted stromal MEs were Stromal and Immune-enriched stromal. Intratumoral TIL (iTIL) and stromal TIL (sTIL) therefore measured lymphocyte density within model-defined compartments rather than whole-tissue lymphocyte abundance (Fig. 4d).

Predicted-ME-derived iTIL was higher in MSI-H than MSS tumors, with a patient-level mean difference of 0.03157. An exact two-sided permutation over all $\binom{10}{4} = 210$ patient-label assignments gave $p = 0.00952$ (Fig. 4e). This direction is consistent with the immune-infiltrated phenotype of MSI-H CRC and shows that predicted compartments can localize lymphocytes to tumor regions. In contrast, sTIL did not differ by recurrence status within either MSS ($p = 0.8333$) or MSI-H ($p = 0.6667$) strata (Fig. 4f). Retaining this negative result is important because only one MSS patient was recurrence free.

Finally, among the six MSS test patients, an outcome-blind Otsu threshold on predicted sTIL divided three patients into each of high- and low-sTIL groups. Median overall survival was 1,993 versus 405 days, respectively, and the exploratory log-rank test yielded $\chi^2 = 5.052$, $p = 0.0246$ (Fig. 4g). The very small groups preclude a validated prognostic claim. Instead, the analysis demonstrates that model-assigned compartments can be substituted into an established clinical workflow and generate testable spatial biomarkers.

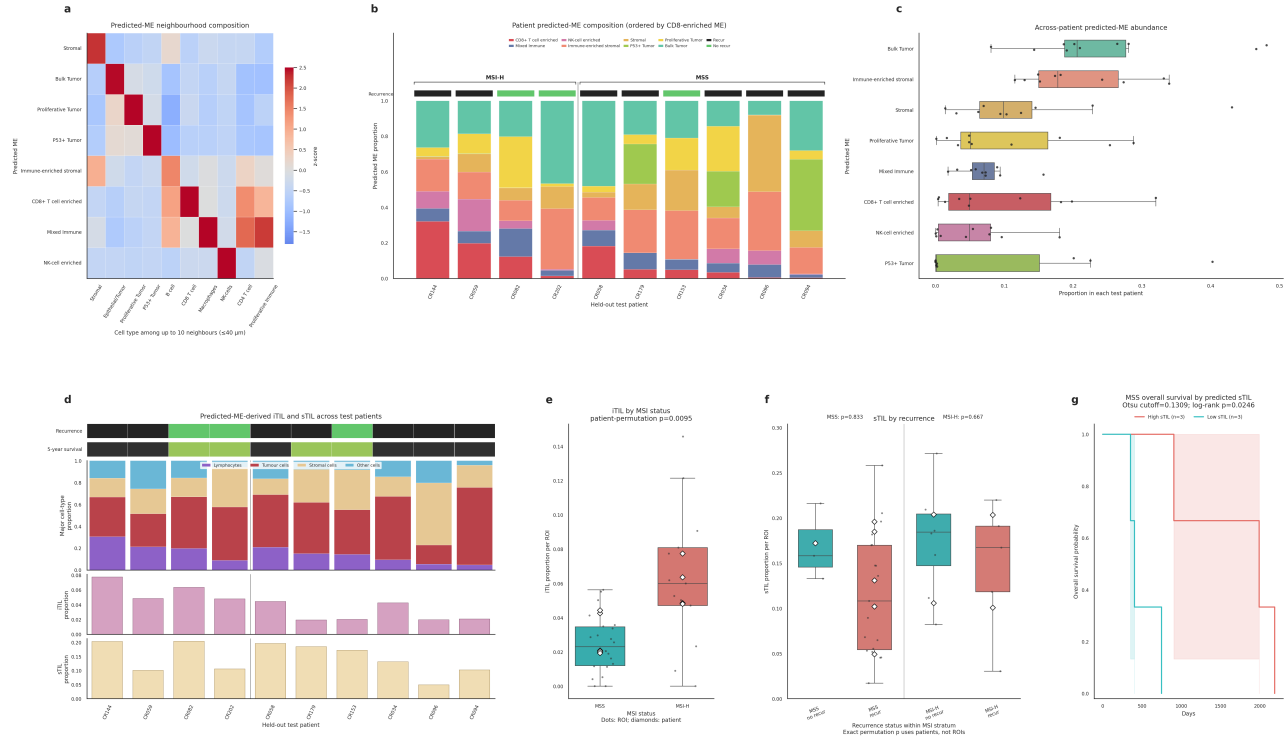


Figure 4: Biological and clinical analyses reconstructed using predicted MEs in held-out CRC patients. A CELLME model trained on 18 patients and selected using ten validation patients was frozen and applied once to ten disjoint test patients. Analyses used 168,295 cells from 35 cancer ROIs. **a**, Column-wise z-scored local cell-type composition of predicted MEs, calculated using up to ten non-self neighbors within 40 μm . **b**, Patient-level predicted-ME composition, stratified by MSI status and ordered by CD8-enriched ME. **c**, Across-patient predicted-ME abundance; points denote patients. **d**, Major cell-type composition and predicted-ME-derived intratumoral and stromal TIL proportions. **e**, Intratumoral TIL by MSI status (exact patient-label permutation $p = 0.0095$); gray points are ROIs and diamonds are patients. **f**, Stromal TIL by recurrence within MSS and MSI-H tumors; neither comparison was significant. **g**, Exploratory overall survival in six MSS test patients stratified by an outcome-blind Otsu cutoff on predicted sTIL (three high and three low; log-rank $p = 0.0246$).

3.4 Slice-level spatial and biological interpretation

Experimental setting. We next examined a single held-out CRC ROI to connect population-level performance to tissue morphology. The ROI was selected from all 35 cancer ROIs in the split-00 test set by a deterministic rule that first maximized the number of predicted MEs with at least 3% abundance, followed by normalized abundance entropy, homotypic predicted-ME edge fraction, and slice identifier. The rule did not use native-label agreement, protein markers, MSI, recurrence, survival, or subjective visual selection. CR034-a2 was the only ROI containing all eight predicted MEs above 3% and comprised 4,387 cells (Fig. 5).

Results and interpretation. The mean maximum-class probability was 0.9848 and mean normalized prediction entropy was 0.0223. Predicted MEs formed contiguous domains: 68.48% of undirected edges in the slice-level 6NN union graph connected cells with the same predicted ME. The full cell-type map and predicted-ME map showed corresponding epithelial, stromal, proliferative, p53-positive, and immune regions (Fig. 5a,b).

Raw local compositions further supported these assignments (Fig. 5c). The Stromal ME contained 78.3%

stromal neighbors; Bulk Tumor contained 60.8% epithelial/tumor neighbors; Proliferative Tumor contained 42.8% proliferative tumor neighbors; P53+ Tumor contained 44.2% p53-positive tumor neighbors; and the NK-cell-enriched ME contained 47.7% NK neighbors. Immune-enriched stromal and Mixed Immune MEs contained mixtures of stromal, T-cell, macrophage, and NK populations, confirming that the predicted ME was not simply a relabeling of the center cell type.

Protein expression was never provided to CELLME. After prediction, marker programs were calculated by within-slice z -scoring and comparing each predicted ME with all other cells in the same ROI. Stromal, epithelial, proliferation, and p53 programs were enriched in the correspondingly named compartments by +1.16, +1.11, +1.38, and +1.26 standardized units, respectively (Fig. 5d). The CD8-enriched ME showed cytotoxic (+0.88) and broad immune (+0.77) enrichment, whereas Mixed Immune showed macrophage (+1.13), B-cell (+0.65), and broad immune (+0.59) enrichment. This is a post hoc coherence check rather than a fully independent validation because upstream cell-type annotations were themselves marker informed.

Uncertainty localized preferentially to ecological transitions. A cell was defined as a boundary cell when at least one 6NN-union edge connected it to a different predicted ME. Mean normalized entropy was 0.02891 for boundary cells and 0.003979 for interior cells, a 7.27-fold ratio; entropy classified boundary versus interior cells with AUROC 0.6375. The deterministic inset was centered on the densest region of top-decile entropy cells, illustrating interdigitating MEs at a tissue transition rather than random instability within homogeneous domains.

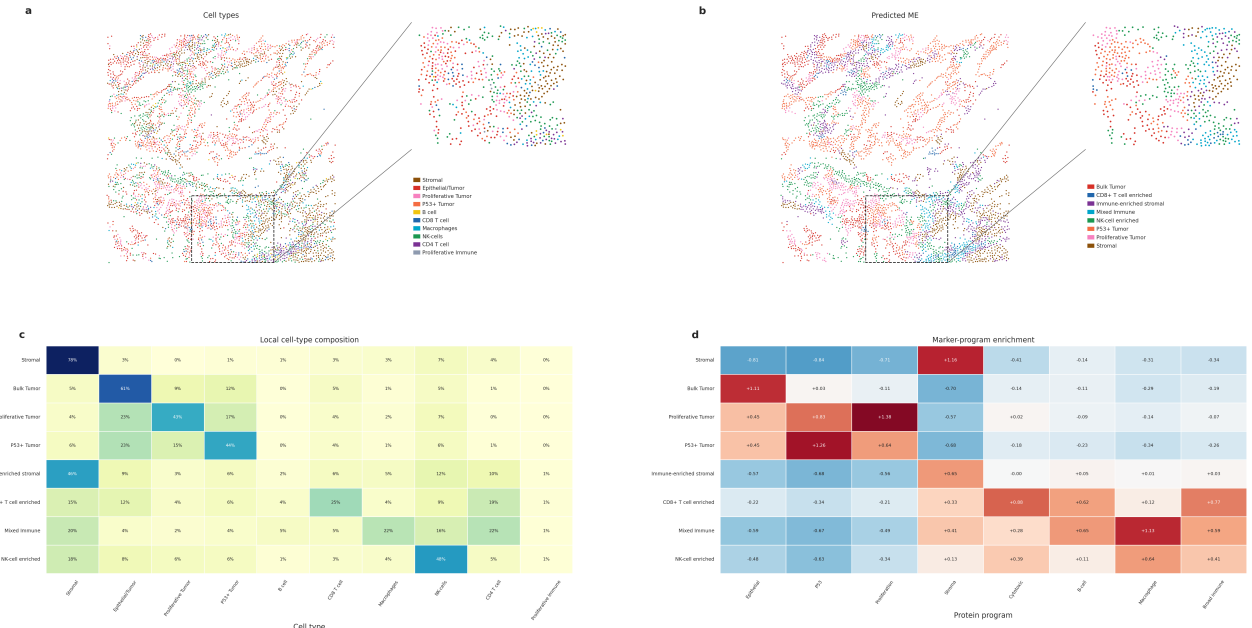


Figure 5: Spatial and molecular coherence of predicted MEs in a held-out CRC tissue ROI. CR034-a2 was selected from 35 test cancer ROIs using a marker-, outcome-, and native-label-blind rule prioritizing the number of predicted MEs with at least 3% abundance. The ROI contained 4,387 cells and all eight predicted MEs. **a**, Spatial cell-type map and deterministic high-entropy crop. **b**, Predicted hard ME labels at the same coordinates and within the same crop. **c**, Raw local cell-type composition of each predicted ME, calculated using up to ten non-self neighbors within 40 μ m; values are percentages and are not z -scored. **d**, Post hoc protein-program enrichment. Markers were z -scored within the slice, and each value is the mean program score in the indicated ME minus the mean in all other cells. Protein measurements were not model inputs. Mean prediction confidence was 0.985, 68.5% of 6NN edges were homotypic for predicted ME, and normalized entropy was 7.3-fold higher at ME boundaries than in domain interiors.

4 Conclusion

This study addresses a recurring inefficiency in spatial proteomics: expert-defined cellular neighborhoods are biologically rich but are commonly rebuilt as cohort-specific products. Native CN discovery is not purely manual—its central clustering step is typically unsupervised—yet substantial expertise enters through phenotype consolidation, neighborhood scale, cluster number, quality control, and biological naming. CELLME preserves that prior work by learning a patient-inductive map from semantic cell types and local spatial organization to the established ontology.

The results distinguish three forms of reuse. First, within a cohort, CELLME labels cells in entirely unseen patients without repeating unsupervised clustering or cluster naming. This transfer was nearly exact for the CRC ontology and consistently strong for NSCLC. Second, a frozen representation retained meaningful organization in an independent CODEX cohort even when the source and target ontologies did not match. Zero-shot ARI and NMI should be interpreted as partition alignment, not target-name accuracy. Third, a simple target-linear head trained on only seven patients aligned the frozen representation to a new nine-class ontology. This low-shot setting is the clearest realization of reusable expert knowledge: the initial source annotation is encoded in the representation, while limited target annotation adapts it to the new study’s vocabulary.

Center specificity and multiscale context were both important. A conventional graph-level pooling operation asks which nodes are generally informative, whereas CenterQuery attention asks which nodes are informative for the ME of a particular center. The same physical cell can therefore receive different relevance depending on the selected center and graph hop. The context branch adds density-adaptive fields of view through nested cell counts rather than a single investigator-chosen radius. Its zero-initialized residual protects the local representation at the start of optimization and lets the model use broader context only when supported by the data. The large CRC and NSCLC improvements over the local CenterQuery model empirically support this design.

The DLBCL results provide an informative counterexample to a universal superiority claim. XGBoost on local phenotype composition was slightly stronger on average and markedly less variable, while CELLME ranked first only at the two largest training proportions. The source DLBCL RCNs were constructed from phenotype-composition motifs, which may make a composition learner especially well matched to the reference labels. In addition, large split-to-split variation suggests patient heterogeneity that is not resolved by adding architectural complexity alone. Future work should evaluate calibration, hierarchical phenotype alignment, and domain-robust training in larger lymphoma cohorts.

Predicted MEs also retained utility beyond label metrics. In held-out CRC patients, model-defined compartments reproduced expected local compositions and enabled the source study’s TIL calculations. The MSI-associated iTIL difference was significant under a patient-level exact permutation, whereas recurrence-associated sTIL comparisons were negative. The exploratory survival split in six MSS patients is hypothesis generating only and requires validation in an independent, adequately powered cohort. Similarly, protein-program agreement in CR034-a2 is encouraging but not entirely independent because protein markers contributed to the upstream semantic cell types. These qualifications are essential to avoid converting ontology agreement into an overstated biological discovery claim.

Several limitations remain. CELLME requires an initial labeled ontology and therefore does not replace the first round of expert discovery; it amortizes that effort across later patients and datasets. Its predictions inherit ambiguity, bias, and granularity from the native labels. The current inputs require harmonized semantic cell types, and the cross-domain adapter was manually specified from names and biological meaning. Automated ontology alignment, open-set detection for unseen cell types or MEs, and uncertainty-aware abstention will be important for deployment. The model uses relative graph

structure and neighbor rank but discards physical distances after neighborhood construction; incorporating calibrated distance features may help when tissue density changes systematically between platforms. Finally, the target transfer experiment used one independent CODEX cohort, and clinical analyses involved a small held-out subset.

Despite these limitations, the experiments establish a practical transition from one-off neighborhood annotation to reusable spatial modeling. A curated ME ontology can be learned from a small patient subset, propagated to unseen patients, transferred as a frozen representation across platforms and cancers, and inserted into downstream biological analyses. This framing complements de novo neighborhood discovery: experts still define and interpret the reference concept, while CELLME makes that concept scalable, reproducible, and portable.

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